SCHOOL OF ADVANCED TECHNOLOGY

SAT301 FINAL YEAR PROJECT

Design and Implementation of a High-Precision Three-Dimensional Force Sensor Based on Triboelectric and Hall Sensors

Final Thesis

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Engineering

Student Name	:	Pengbo Gao
Student ID	:	2252496
Supervisor	:	Chun Zhao
Assessor	:	Yuxiao Fang

Abstract

With the development of robot tactile perception technology, tactile sensors not only need to sense the magnitude of external forces, but also need to have the ability to identify the material of the contact. This project proposes a dual-modal tactile sensing scheme that combines Hall sensors with triboelectric nanogenerators (TENGs) to achieve three-dimensional force detection and material identification.

In the three-dimensional force detection section, this research designed and fabricated a Hall sensor based on the MLX90393 Hall sensor chip. When an external force is applied, the force is transmitted through the PDMS pressing layer, causing the embedded magnet to shift relative to the device chip, thereby resulting in a change in the magnetic induction intensity. The test results show that this module can generate incremental outputs of magnetic induction intensity that vary with the external force in the x, y, and z directions, proving the feasibility of achieving three-dimensional force detection based on magnetic field changes.

In the material identification section, this project fabricated a test device based on TENG and used the single-electrode testing method to test various materials. The test results showed that different materials could generate frictional electric output signals with significant differences, mainly manifested in the voltage amplitude and signal polarity, proving the feasibility of using TENG for material identification.

Overall, this project successfully fabricated and tested the Hall three-dimensional force detection device and the TENG material recognition device, verifying the feasibility of their application in dual-modal tactile perception. This provided an experimental basis for the subsequent functional integration and array design of the devices.

Keywords: Hall sensor; Triboelectric nanogenerators; Three-dimensional force detection; Material identification

Acknowledgements

First of all, I would like to express my sincere gratitude to Professor Chun Zhao for the guidance and support he has provided throughout this project. From determining the research direction and designing the experimental protocols to analyzing the results and drafting the report, my supervisor has offered invaluable advice, enabling me to gain a more comprehensive understanding of the research involved and to successfully complete each stage of the work.

Secondly, I would like to thank Dr. Hao Lei for his assistance and guidance throughout the project. Whether in device fabrication, testing methods or the analysis of experimental issues, Dr. Hao Lei provided me with a great deal of support, enabling me to better resolve the problems encountered during the course of the project.

Furthermore, I would like to express my gratitude to the university and the laboratory for providing the experimental facilities, equipment and learning environment, which have been instrumental in ensuring the smooth progress of this project.

Finally, I would like to thank my family and friends for their understanding, support and encouragement throughout my undergraduate studies. This experience has been of great significance to my learning and personal development.

Contents

List of Tables

List of Figures

List of Acronyms

Term	Initial components of the term
Al	Aluminium
Cu	Copper
DI water	Deionized Water
FA	Fast Adapting
FEP	Fluorinated Ethylene Propylene
IPA	Isopropyl Alcohol
KH560	γ-Glycidoxypropyltrimethoxysilane
PCB	Printed Circuit Board
PDMS	Polydimethylsiloxane
PLA	Polylactic Acid
PMMA	Polymethyl Methacrylate
PTFE	Polytetrafluoroethylene
PU	Polyurethane
SA	Slowly Adapting
TENG	Triboelectric Nanogenerator
TPU	Thermoplastic Polyurethane

Chapter 1

Introduction

1.1 Motivation, Aims and Objective

With the rapid development of robot technology and intelligent systems, achieving dexterous manipulation at a human-like level has become an important research direction. However, in complex manipulation tasks, relying solely on simple force feedback is no longer sufficient; the key lies in the multi-dimensional information support provided by tactile perception.

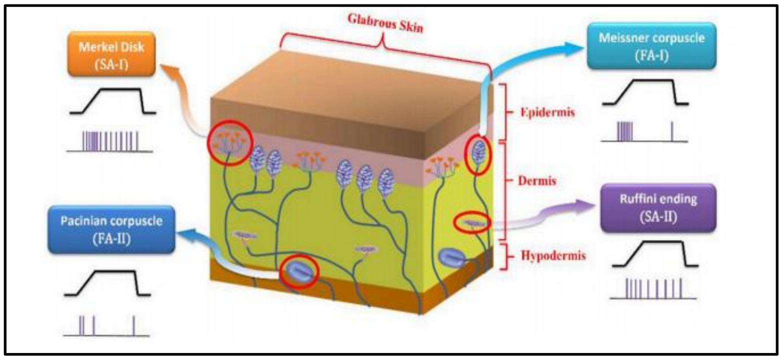


Figure 1.1: Mechanical receptors under human skin [1]

From a biological perspective, humans can achieve highly accurate perception of the external environment through the mechanical receptors in their skin, thereby enabling complex and precise operations. As shown in Figure1.1, tactile perception mainly relies on two types of mechanical receptors: slow adaptation (SA) and fast adaptation (FA). The SA receptors generate responses under continuous mechanical stimulation, and are mainly used to perceive static pressure and shape information; while the FA receptors mainly generate responses at the beginning or during changes of the stimulus, and are more sensitive to dynamic information such as vibration and sliding [1]. The two transmit through different channels and integrate in the nervous system, thereby achieving the comprehensive perception of complex tactile information. This perception mechanism provides important inspiration for the design of bionic tactile sensors.

In practical applications, such as robot grasping, not only the magnitude of the contact force needs to be obtained, but also the direction of the force and the material properties of the contacting object need to be perceived. Research shows that three-dimensional force information is crucial for the stability of grasping, especially when dealing with fragile objects, where the magnitude and direction of the contact force directly determine the success of the task [2]. At the same time, the material recognition ability can help the system further determine the characteristics of the object, thereby optimizing the control strategy.

However, most of the existing tactile sensors focus on the realization of a single function. For instance, tactile sensors based on optical fibers can achieve high-sensitivity pressure and vibration detection, but due to their single sensing unit, they cannot achieve three-dimensional force perception; methods based on vision and neural networks can construct three-dimensional force distribution, but they are usually complex in structure and have low system integration [3]. Additionally, these approaches mainly focus on the acquisition of force information and lacks the perception of material properties, making it difficult to achieve the collaborative perception function of static and dynamic information in the human tactile system.

In response to the above issues, this paper proposes a dual-modal tactile sensor based on the combination of Hall effect and triboelectric nanogenerator. This design mimics the collaborative mechanism of SA and FA receptors in the human tactile system: using the Hall sensor (corresponding to SA) to achieve three-dimensional force detection; at the same time, using TENG (corresponding to FA) to achieve material identification, and distinguishing different materials through the characteristic electrical signals generated during their contact process.

The core objective of this research is to develop a simple-structured and clearly-functioning dual-modal tactile sensor that can effectively integrate three-dimensional force detection and material identification. As a first step towards achieving this goal, the specific project contents include:

- Design a three-dimensional force detection structure based on magnetic field variation feedback to achieve precise measurement of forces in the X, Y, and Z directions.

- Construct a material identification layer based on TENG to achieve the distinction of different materials' electrical signals.
- Verify the feasibility of the three-dimensional force detection and material identification functions through experiments.

1.2 Literature Review

In recent years, the field of tactile sensing has been evolving from the detection of a single physical quantity to multi-modal perception. To simulate the functions of the human tactile system, researchers have been constantly attempting to integrate various tactile information such as force, vibration, and material properties into the same sensing system. Similar to the cooperative perception of slow adaptation (SA) and fast adaptation (FA) receptors in the human sensory system, multimodal tactile sensors usually need to simultaneously acquire different types of tactile signals. However, achieving the acquisition of multi-modal information still faces significant challenges.

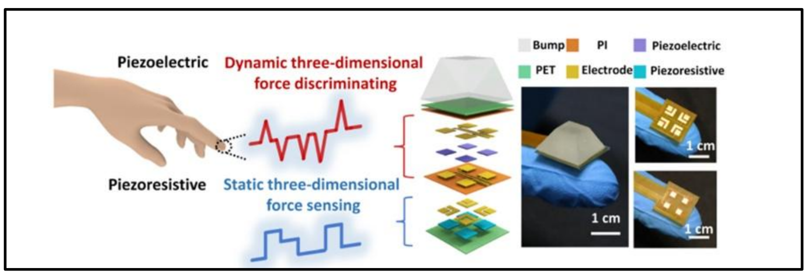


Figure 1.2: Bimodal haptic sensor [4]

In the field of three-dimensional force detection, existing studies have shown that the acquisition of multi-directional forces is of great significance for the grasping operations of robots. The current three-dimensional force sensors are usually implemented based on the piezoelectric + piezoresistive mode, as shown in Figure 1.2, this sensor adopts a layered structure design. On the flexible substrate, the piezoelectric layer and the piezoresistive layer are respectively constructed, and the external three-dimensional force is effectively transmitted to the sensor interior through the top protruding structure. Among them, the piezoelectric layer is mainly used to respond to high-frequency dynamic stimulation, while the piezoresistive layer is used to detect continuous static pressure. From the figure, it can be seen that the piezoelectric layer and the piezoresistive layer are respectively arranged with 4 electrode array units. This design can ensure that under the normal force, the responses of each electrode unit are uniformly distributed, and when the tangential force exists, it presents obvious asymmetric characteristics. However, this method generally has problems such as complex structure and temperature sensitivity. Moreover, due to the signal superposition among different electrode array units, complex decoupling algorithms are usually required. In addition to the above, in order to further enhance the tactile perception ability, some research based on traditional tactile sensors has also attempted to incorporate torque and twist angle into the measurement range by utilizing the differences in multiple components' signals. However, there are still deficiencies in terms of resolution, linear range, and signal decoupling [5]. Therefore, how to reduce the structural complexity while maintaining performance remains a key issue in current sensor design.

In terms of material identification, various identification methods have been proposed in previous studies, including those based on thermal conductivity, ultrasonic waves, and computer vision, etc. However, these methods each have their own drawbacks, such as long response times, limited applicability, or high system complexity [6]. In contrast, TENG is capable of effectively distinguishing different materials and has a high level of recognition accuracy. Triboelectricity is a technology with significant potential for application in energy harvesting and self-powered systems. Its fundamental principle involves converting mechanical energy into electrical energy, thereby providing the power required to drive microelectronic devices. TENGs developed on the basis of this principle generate an electrical signal through the transfer of charge caused by surface friction when two different materials come into contact, and through the formation of a potential difference between the electrodes via electrostatic induction as the materials separate [7].

Therefore, this paper proposes a dual-modal tactile sensing scheme based on the combination of Hall sensors and TENG. By applying the Hall sensor for three-dimensional force detection and the TENG for material recognition, the functions of different sensing mechanisms are separated, thereby avoiding the signal superposition and complex decoupling processes in traditional methods. While ensuring a relatively simple system structure, this approach achieves an enhancement in multi-modal tactile perception capabilities.

Chapter 2

Methodology and Results

2.1 Methodology

This study proposes a dual-modal tactile sensor based on the combination of Hall effect and triboelectric effect. The overall structural concept design is shown in Figure 2.1:

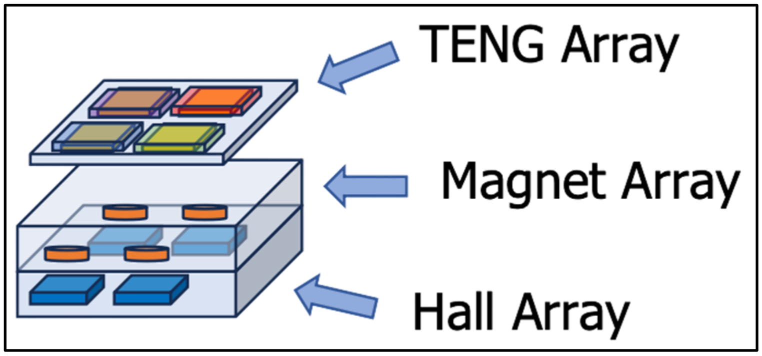


Figure 2.1: Structural design

This system mainly consists of a TENG array layer, a magnet array layer, and a Hall sensor array layer. Among them, the top TENG layer is used for material identification, the middle magnet array is used to convert external force stimulation into flux changes, and the bottom Hall sensor array is used to detect flux density changes, thereby achieving three-dimensional force measurement. This design is inspired by the collaborative working mechanism of SA and FA mechanoreceptors in human skin. By separately assigning different perception tasks to different physical mechanisms, this study effectively avoids the common problems of signal superposition and complex decoupling that exist in traditional multimodal tactile sensors.

It should be noted that although the design shown in Figure 2.1 is a 2×2 array structure, its main purpose is to illustrate the expansion direction of future multi-unit integrated tactile sensing systems. During this research stage, only three-dimensional force devices and frictional electrical devices with a single sensor were fabricated and tested to verify the feasibility of this scheme.

2.1.1 Hall sensor design

The three-dimensional force detection module in this project is designed based on the commercial MLX90393 Hall sensor chip. The MLX90393 is a high-sensitivity three-axis Hall sensor that can achieve real-time detection of magnetic flux density changes, making it suitable for sensing tiny displacements and multi-directional external forces. Although the Hall sensor chip is a commercial component, in this paper, to meet the requirements of three-dimensional force detection, the overall structure of the device has been redesigned, including the sensor shape, layout, and interface positions.

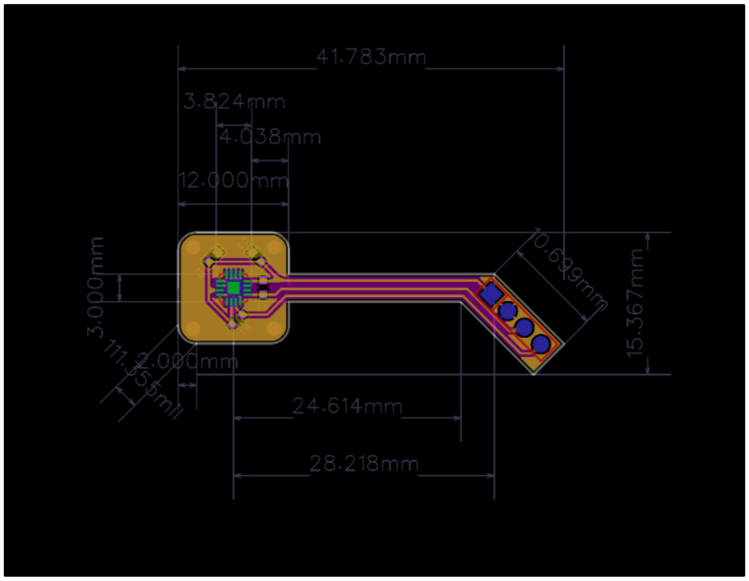
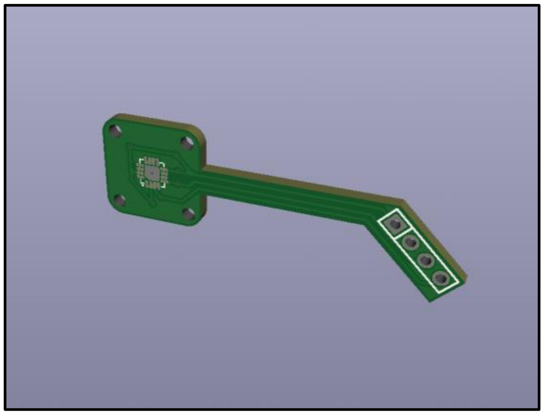


Figure 2.2: Hall sensing PCB layout design



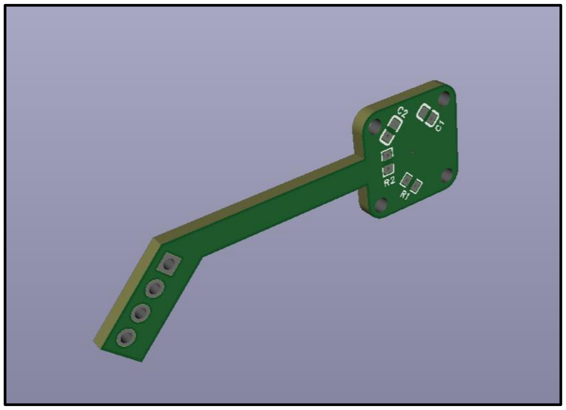


Figure 2.3: 3D structural view of the Hall sensing

As shown in Figure 2.2, the entire PCB adopts a two-layer structure, with a board thickness of 1.6mm and a total size of approximately 41.8mm × 15.4mm. The main body can be divided into three parts: the Hall chip area, the signal transmission area, and the external interface. Among them, the area of the Hall chip on the left measures approximately 12mm × 12mm and the MLX90393 chip is placed in the center for detecting the magnetic field changes caused by the movement of the magnet. The middle part adopts a narrow and long transmission structure. By increasing the distance between the sensing area and the external interface, the influence of external vibrations and pulls on the measurement results can be reduced. The right area is the external interface, used for power input and signal output connection and this part adopts an inclined layout design. From a three-dimensional perspective (Figure 2.3), the compact layout structure of the Hall sensing module can be observed more intuitively. This design not only keeps the overall structure miniaturized but also effectively reduces the impact of external mechanical disturbances on the sensing detection.

After completing the fabrication of the aforementioned Hall sensor components, they need to be encapsulated to enhance the overall stability of the devices. Since the MLX90393 Hall sensor has a high sensitivity to magnetic field changes, external mechanical disturbances or local displacement of the device may affect the measurement results. Therefore, encapsulating the Hall device is of great significance for ensuring the accuracy of three-dimensional force detection. Considering that epoxy resin possesses excellent mechanical strength, electrical insulation properties, chemical stability and good molding capabilities, this research adopts it as the material for device packaging. It is worth noting that, considering the stability of the device packaging process, this study also used 3D printing to create auxiliary packaging molds and device fixation structures. As shown in Figure 2.4, the epoxy resin packaging mold was printed using TPU85A material, while the device handle fixation structure was printed using PLA material. The reason for using TPU85A material is

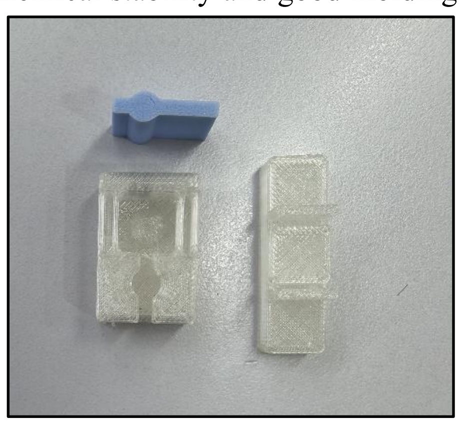


Figure 2.4: The TPU85A and PLA structures

that it has certain flexibility and elasticity, which greatly enhances the convenience of demolding. Moreover, the flexibility of this material can also adapt to the minor deformations during the epoxy resin curing process to some extent, thereby improving the packaging integrity. Before pouring the epoxy resin, in order to enhance the ease of demolding for the packaged devices, this study uniformly sprayed a small amount of demolding agent on the inner wall of the mold. The demolding agent can form an isolation layer between the mold and the epoxy resin, thereby reducing the adhesion between the epoxy resin and the mold after curing. After the excess demolding agent has evaporated, the Hall device is placed in the mold and the uniformly mixed epoxy resin AB glue is slowly injected. The device after the mold filling is shown in Figure 2.5. Subsequently, the device is placed in a room temperature environment for preliminary curing for 4 hours, and then further heated and cured at a constant temperature of 60°C for 2 hours. The packaged device is as follows:

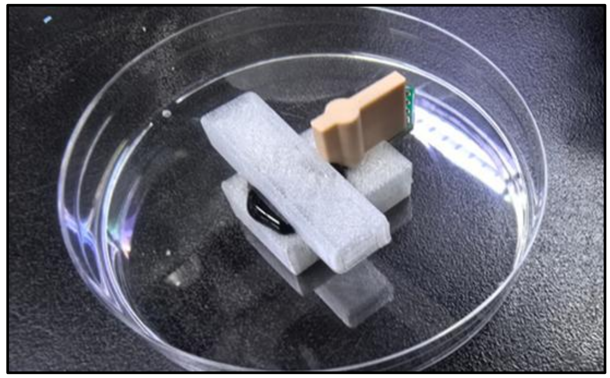


Figure 2.5: Packaging process

Figure 2.6: Hall senser with epoxy resin packaging

After the epoxy resin packaging is completed, the pressing layer needs to be constructed. Considering that Polydimethylsiloxane (PDMS) has a relatively low elastic modulus, good flexibility and elastic deformation ability, this project selects it as the pressing layer in the Hall three-dimensional force detection device. However, there is inherently poor interfacial adhesion between PDMS and epoxy resin, which easily leads to interface delamination during subsequent pressing and repetitive experiments. Moreover, previous studies have indicated that due to the limited interfacial compatibility between silicone rubber and epoxy resin, the adhesion issue often limits the practical application of the composite structure [8]. Therefore, in order to ensure the stability and durability of the device, this research conducted interface modification treatment on the surface of the epoxy resin before PDMS coverage. During the processing, the KH560 silane coupling agent will be used to modify the surface of the epoxy resin. Previous studies have shown that KH560 can be used as a co-crosslinking agent between PDMS and epoxy resin. After modification with KH560, a high bonding strength interface can be formed between PDMS and epoxy resin. The processing procedure is as follows:

1. Using 600-mesh sandpaper, the upper surface of the packaged device was polished to increase the surface roughness. Subsequently, the surface was cleaned successively with isopropyl alcohol (IPA) and deionized water to remove the particles and contaminants remaining from the polishing process. After cleaning, the device was placed in a 60°C environment for 10 minutes of drying treatment.
2. The surface of the epoxy resin was treated with a 2% NaOH solution for alkaline activation. This was done to form more active hydroxyl groups on the material surface, facilitating the subsequent bonding of the silane coupling agent to the epoxy resin surface. After the activation process was completed, the device was rinsed with DI water and thoroughly dried.
3. Prepare the KH560 silane coupling agent solution. In the research, 95:5 volume ratio of anhydrous ethanol and deionized water was mixed, and 2% of γ-Glycidoxypropyltrimethoxysilane (KH560) was added. After the mixture was completed, a small amount of 0.5 mol/L anhydrous citric acid was added to adjust the solution pH to approximately 4 to promote the hydrolysis reaction of KH560. Then, the solution was left to stand for 30 minutes to allow the silane coupling agent to fully pre-hydrolyze.

4. Immerse the alkaline activated device in the pre-hydrolyzed KH560 solution for approximately 1 to 5 minutes, then remove it and let it stand for about 10 to 20 minutes to form a stable coupling layer on the surface.

After the surface modification process is completed, the PDMS pressing layer is then constructed. To ensure the stability of the PDMS filling process, a set of auxiliary molds was designed and fabricated in this study, as shown in Figure 2.7. Furthermore, it can be observed that the interior of the mold cover in the figure is designed with a thin cylindrical structure. This is to reserve space for placing the magnets in the PDMS layer later.

Figure 2.7: 3D-printed mold for PDMS layer fabrication

Before injecting PDMS, it is necessary to stir it evenly. Since bubbles are likely to be introduced during the stirring process, it is necessary to place it in a vacuum chamber for defoaming for about 15 minutes. After degassing is completed, slowly inject PDMS into the mold to evenly cover the surface of the device and fill the mold space. After the injection is finished, place the entire structure in an environment of approximately 60°C and allow it to cure for 80 minutes. After the curing process is completed, the device will look as follows:

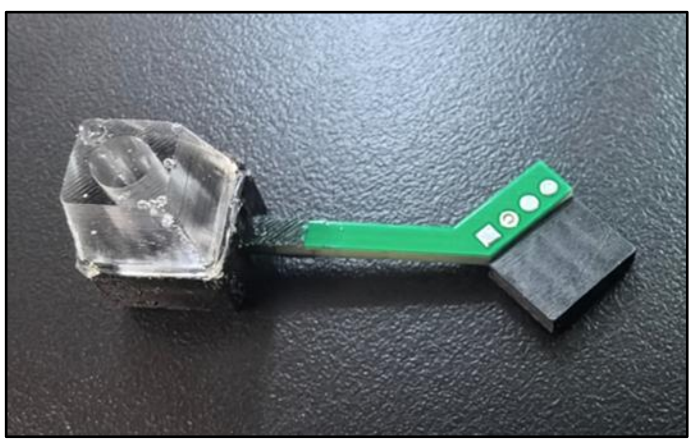


Figure 2.8: Cured PDMS pressing layer

From the above figure, it can be seen that this study adopts a truncated pyramid structure. This structure features a top platform and inclined side walls. It is prone to elastic deformation under small forces, thereby enhancing the pressure response of the sensor. Under larger forces, this structure is less likely to be completely compressed, which helps to expand the measurement range of the device. Furthermore, the flat top surface increases the contact area between the device and external objects, thereby improving the device's stability and robustness when subjected to actual forces [9].

After ensuring that the first layer of PDMS has fully cured, place the magnet in the reserved cylindrical space, and then perform the second PDMS sealing process to enhance the surface uniformity and final packaging integrity. After being heated and cured at a constant temperature of 60°C for 80 minutes, a stable PDMS-Epoxy resin composite structure is finally formed, as shown in the figure below:

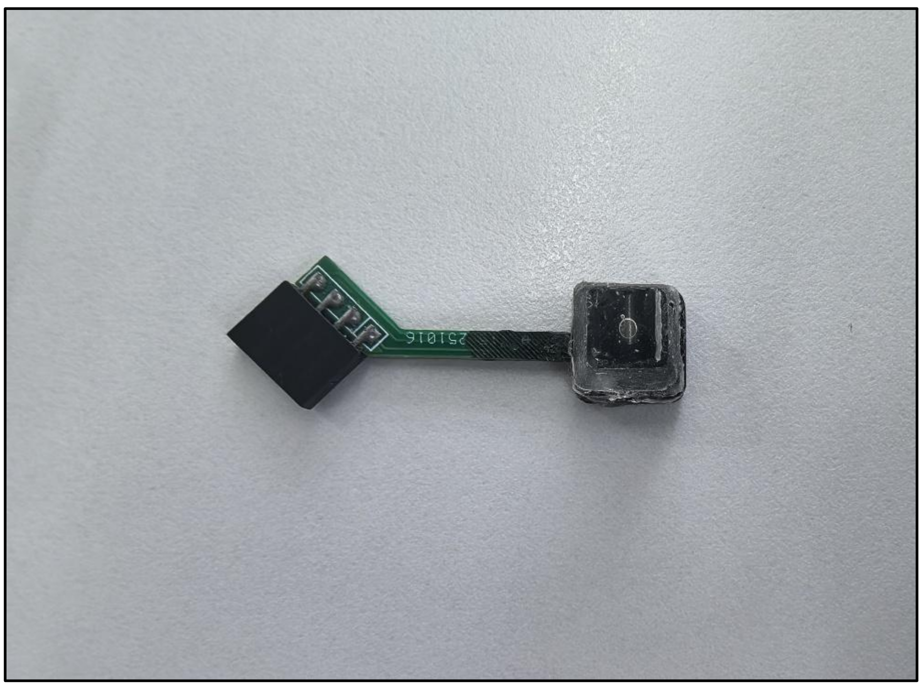


Figure 2.9: Final Hall sensing device

2.1.2 TENG design

After completing the fabrication of the three-dimensional force detection module, this research will further construct a material identification module based on TENG, which is used to obtain the frictional electrical signals generated when contacting different materials. Previous studies have indicated that the working mechanism of TENG is a combination of contact electrification and electrostatic induction. Contact electrification is used to generate static polarized charges on the material surface, while electrostatic induction is responsible for the main mechanism for converting mechanical motion into electrical signals [10]. When two materials with different electron affinity properties come into contact and separate, positive and negative charges will be generated on their surfaces respectively; when the charged materials slide relative to each other, an electric current will be generated in the external circuit, thereby forming measurable electrical signals. Due to the differences in triboelectric properties of different materials, the output electrical signals will also show significant differences in amplitude and polarity, thus being applicable for material identification.

The TENG module used in this research is mainly composed of a silicone contact layer, an aluminum electrode layer, a copper wire, and an acrylic base, as shown in Figure 2.10. It is worth noting that to enhance the frictional electric output during the contact and separation process, micro-structural protrusions were designed on the surface of the silicone contact layer. Existing studies have shown that these micro-structural protrusions can increase the overall contact area between the TENG and the contacting object, thereby improving the output performance [11]. Additionally, the aluminum layer serves as the conductive layer, used to collect the charges generated during the frictional process; the copper wire is used to lead the electrical signals to external testing equipment; and the acrylic layer acts as the base of

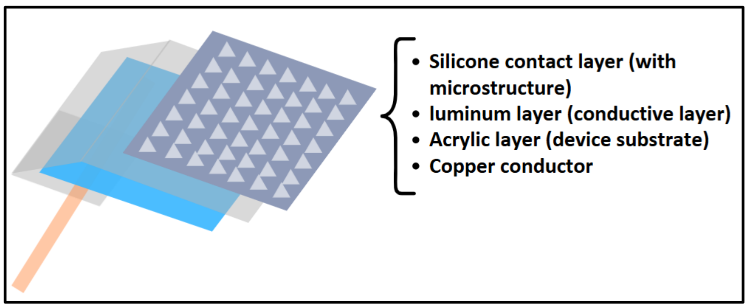


Figure 2.10: Layered structure of the TENG device

the device, providing mechanical support for the overall structure. This layered structure can achieve stable frictional electric signal collection while maintaining a simple device structure. The physical device is as follows:

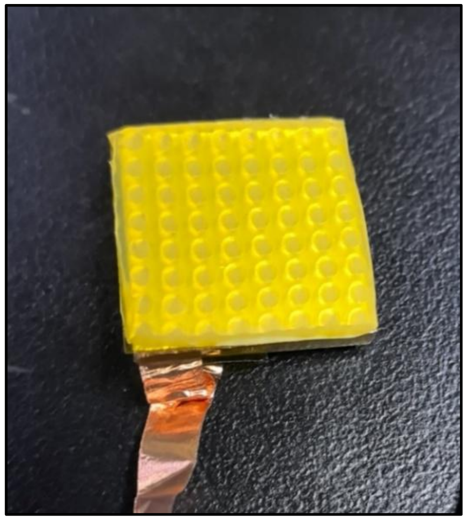


Figure 2.11: Fabricated TENG device

After the TENG device (the lower electrode plate) was fabricated, in order to achieve the identification function for different materials, this study further prepared various upper electrode plate contact materials as test objects, as shown in Figure 2.12.

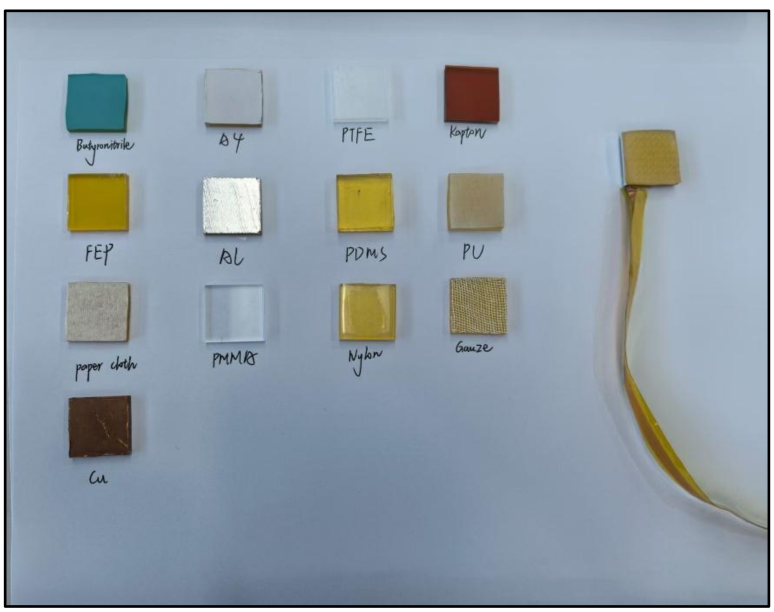


Figure 2.12: Upper plates with different contact materials

The selected materials included nitrile, A4 paper, PTFE, Kapton, FEP, Al, PDMS, PU, paper cloth, PMMA, nylon, gauze, and Cu. These materials have significant differences in terms of conductivity, flexibility, and triboelectric properties. Therefore, when they come into contact and separate with the lower silicone microstructure contact layer, different triboelectric output signals will be generated.

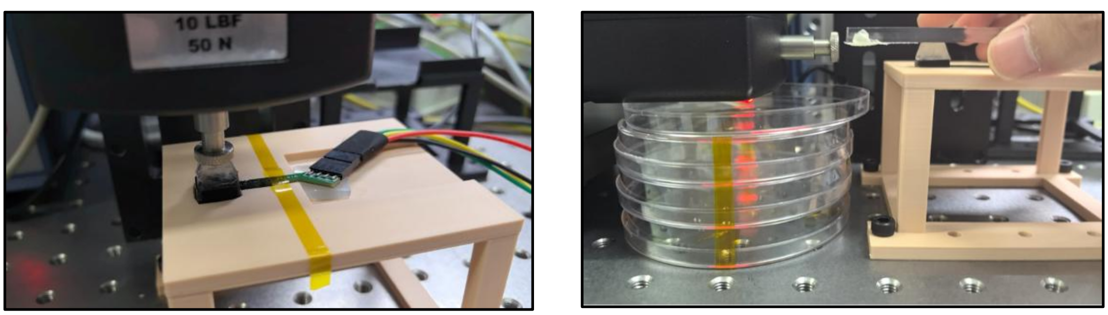
In order to ensure the comparability of the test results for different materials, all the upper electrode contact materials were cut to the same size(1cm×1cm), and during the test, the contact position, contact area and normal force were maintained at the same values. Furthermore, in the subsequent tests of this study, single-electrode mode was adopted for signal acquisition. In this testing method, only the copper wire of the lower electrode plate was connected to the signal acquisition device. Compared with the traditional dual-electrode contact-separation structure, the single-electrode mode does not require additional wires to be connected to the upper contact material. Therefore, it is more suitable for rapid replacement and testing of multiple independent experimental materials. In order to obtain reliable material identification data, each material underwent multiple repeated contact-separation tests, and the corresponding output voltage signals were recorded. In the subsequent analysis, the peak output voltage and the polarity of the signal were mainly selected as the parameters for material identification. By comparing the differences in frictional electrical signals produced by different materials under the same test conditions, the correspondence between material types and output signal characteristics can be established.

2.1.3 Three-dimensional force testing process

Following the fabrication of the aforementioned device, this study tested and calibrated the output response of the three-dimensional force sensor under various external forces. The primary objective of this testing was to establish a quantitative relationship between the external force and the three-axis magnetic field output of the Hall sensor, thereby verifying the feasibility of using this device for three-dimensional force detection.

This study utilized the AMETEK Chatillon DFS II series digital force gauge as the external force application device, as shown in Figure 2.13. This device is capable of applying and displaying external forces, thereby ensuring good controllability and repeatability of the forces applied during the test. During testing, the sensor device was secured to the experimental platform such that the PDMS compression layer was positioned beneath the loading probe. During the loading process, the probe of the force gauge made contact with the surface of the device and gradually applied an external force. In this manner, changes in the Hall sensor's output could be recorded under conditions of known force values.

Figure 2.13: Digital force gauge



A **B**

Figure 2.14: Three-dimensional force testing fixture. (A) Normal force; (B) tangential force

As shown in Figure 2.14, this study has respectively constructed the testing fixtures for normal force and tangential force. Among them, the normal force test is used to detect the Hall output response of the device under the pressure in the z direction; the tangential force test is used to detect the output change of the device under the shear force in the x and y directions. By testing the force conditions in different directions separately, the corresponding relationship between the three-dimensional external force and the three-axis magnetic field signal can be established.

Prior to the formal test, the loading probe is first lowered slowly using the three-axis controller software, bringing it into slight contact with the surface of the PDMS compression layer. At this stage, the probe does not exert any significant pressure on the device; it is used solely to determine the initial contact position. Subsequently, the digital push-pull force gauge is zeroed, establishing this state of slight contact as the reference point for subsequent force loading tests. This minimizes the impact of variations in the probe’s initial position on the test results and ensures that all test groups start under identical conditions.

Once the initial calibration is complete, the three-axis controller software is used to further control the probe’s travel distance in a specified direction, thereby altering the magnitude of the constant force applied to the device. During the testing process, when the PDMS pressing layer is subjected to external force, it will undergo elastic deformation, which in turn drives the internal magnet to shift relative to the Hall sensing chip. Since the Hall sensor can detect changes in the spatial magnetic field distribution, the change in the magnet's position will cause variations in the output of the B_x, B_y and B_z magnetic flux densities. By comparing the changes in the three-axis magnetic field signals under different force conditions, the relationship between the force magnitude, direction, and the Hall sensor's output response can be established.

For the force tests in the x and y directions, this study used an action force interval of 0.5N, that is, each time the force increased by 0.5N, a set of corresponding Hall sensor output data was recorded. The test range was 0–3N. This range was mainly used to characterize the response capability of the device under tangential force. For the normal force test in the z direction, since the normal pressure is more common in the actual tactile detection and pressing interaction scenarios, the test range was expanded to 20N, which was used to evaluate the output response of the device under larger normal force conditions. When reaching a target force value and stabilizing, the corresponding Hall three-axis output signal was recorded. During the test, the Arduino UNO was used to read the data from the three-dimensional force sensor and transmit it to the computer via serial communication and monitored using the SerialPlot software. The software interface is shown in the Figure 2.15.

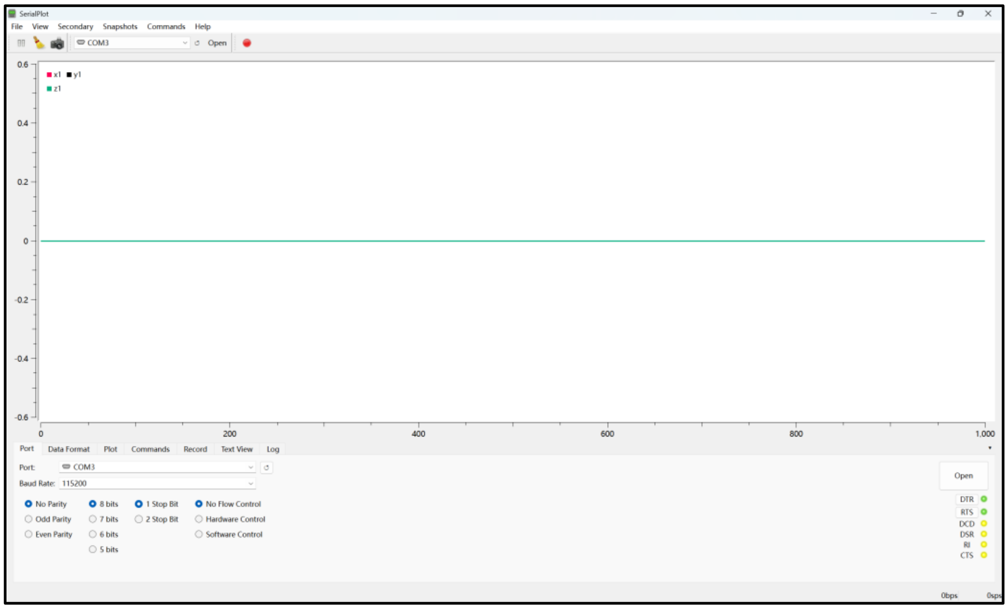


Figure 2.15: The interface of the SerialPlot application

In the SerialPlot interface, three sets of data channels are set up to display the changes in Hall output in the x, y and z directions respectively. Here, the serial communication baud rate is set to 115200, and the data format is 8 data bits, 1 stop bit, no parity check, and no flow control settings. Through the real-time curve display function of SerialPlot, the dynamic changes of the three-axis magnetic field signals during the test can be observed intuitively, and the rationality of the output data can be judged.

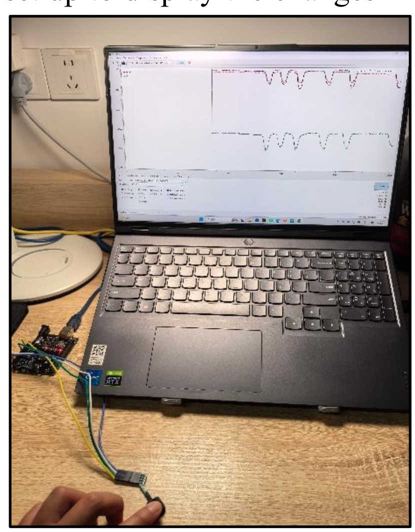


Figure 2.16: Informal testing process

During the testing process, as the applied force gradually increases, the three-axis curves in SerialPlot will change accordingly with the displacement of the magnet, as shown in Figure 2.16. If the testing process is stable, the curves usually show a relatively regular trend of change; if there are situations such as the probe of the force gauge shifting, unstable contact, or device sliding, the output curves may exhibit significant fluctuations or abnormal sudden changes. Therefore, SerialPlot is not only used for data acquisition, but also for real-time judgment of the signal stability during the testing process.

After each group of tests is completed, the data recorded by SerialPlot will be exported as a CSV file and imported into OriginPro software for further processing and fitting analysis. During the data processing, the original data is first sorted, and invalid data segments and obvious outliers are removed; then, the data in the stable stage under each force value is selected as the valid output. For different force application directions, the corresponding changes in B_x, B_y and B_z output are extracted and compared with the force values displayed by the force gauge. Finally, a curve fit is performed in OriginPro to analyze the relationship between the applied force and the Hall output signal, yielding calibration curves for forces in different directions. The results of this fit can be used to further analyze the sensor's sensitivity and linearity in the x, y and z directions.

2.1.4 The triboelectric testing process

After completing the three-dimensional force test, this study further conducted the material identification test. The main purpose of this test was to obtain the triboelectric signals generated during the contact-separation process between different materials and the silicone microstructure contact layer, and to verify the feasibility of using the TENG module for material identification by comparing the differences in corresponding signals of different materials.

The system of this test mainly consists of the lower electrode plate of TENG, replaceable upper contact material, contact-separation device, three-axis motion controller software, and signal acquisition software. Among them, the lower electrode plate of TENG serves as the fixed testing end and is mainly responsible for generating and outputting triboelectric signals; the upper electrode plate has replaceable contact materials, which are used to simulate the signal differences when different materials come into contact with the sensor surface; the AMETEK Chatillon DFS II series digital force gauge was mainly used as a contact-separation device in this experiment, to drive the upper electrode plate to perform up-and-down reciprocating motion. At the same time, the force value display of this gauge was also used to assist in judging the pressing force during the contact process, thereby improving the consistency of different material testing conditions.

The force gauge is controlled by the three-axis controller software, through this software, different movement distances and cycle counts can be set. In this test, the main method is to adjust the descent and ascent of the loading probe, so that the upper electrode can periodically contact and separate from the lower electrode in the same movement pattern. This control method can reduce the instability caused by manual pressing, and ensure that the movement trajectories and contact methods during the testing of different materials remain consistent.

In this test, TENG signals were collected using a single-electrode mode. In this testing method, the lower electrode plate with a micro-structured silicone contact layer served as the main test electrode, and its copper wire was connected to the positive terminal of the signal acquisition end; the upper contact material was not connected to any wires and was only used as a replaceable friction contact material. At the same time, both the negative terminal and the ground terminal in the testing system were connected to the ground wire. Compared with the traditional double-electrode contact-separation structure, the single-electrode test does not require additional wires to be connected to the upper electrode plate, thus being more suitable for rapid replacement of various independent test materials.

Before the formal test, the initial contact position calibration needs to be carried out first. Specifically, the loading probe is controlled to slowly descend through the three-axis controller software, allowing the upper electrode to gradually approach the contact layer of the lower TENG microstructure. When the upper contact material is in a slight contact state with the surface of the lower electrode, the reading of the AMETEK Chatillon DFS II force gauge is set to zero, and this position is taken as the initial position for the subsequent contact-separation test. This step ensures that there are relatively consistent initial contact conditions during tests with different materials, reducing the signal differences caused by different initial spacings or contact states.

After the calibration is completed, the preset contact-separation motion program is executed through the three-axis controller. According to the parameter settings, each material completes 5 consecutive contact-separation rounds in one test. In each cycle, the upper electrode plate first moves downward and contacts the silicone microstructure layer, then moves upward and leaves the contact surface. To further ensure the consistency of test conditions for different materials, this test uniformly requires: after the upper electrode plate descends and contacts the lower electrode plate, the pressure reading should reach approximately 5 N before the upward separation is carried out. If the actual test process shows that the pressure is significantly higher or lower than 5 N, the movement distance parameter in the three-axis controller is adjusted to make corrections, so as to ensure that the contact force and contact process for different material tests are as consistent as possible.

In addition, in order to enhance the comparability of the test results, all the contact materials were cut to the same size, namely 1cm×1cm. During the testing process, efforts were made to maintain the same contact position, contact area, initial contact state, motion mode, and pressure conditions. By using this method of controlling variables, the errors introduced by changes in test conditions can be minimized, and the differences in the output signals mainly result from the triboelectric properties of the materials themselves.

During a complete contact-separation cycle, when the upper electrode comes into contact with the lower electrode, the interface between the two materials generates frictional charges due to contact electrification. When the upper electrode moves upward and separates from the lower electrode, the distance between the two materials changes, causing a shift in the electric potential distribution. This change in potential is then driven by electrostatic induction to cause the movement of electrons in the external circuit, thereby generating a voltage signal. Due to the differences in the ability to gain or lose electrons, the degree of softness, and the conductivity of the two materials, the corresponding output signals will exhibit different characteristics in terms of voltage amplitude and polarity.

The output signals are monitored and recorded in real time using the LabVIEW software, as shown in Figure 2.17.

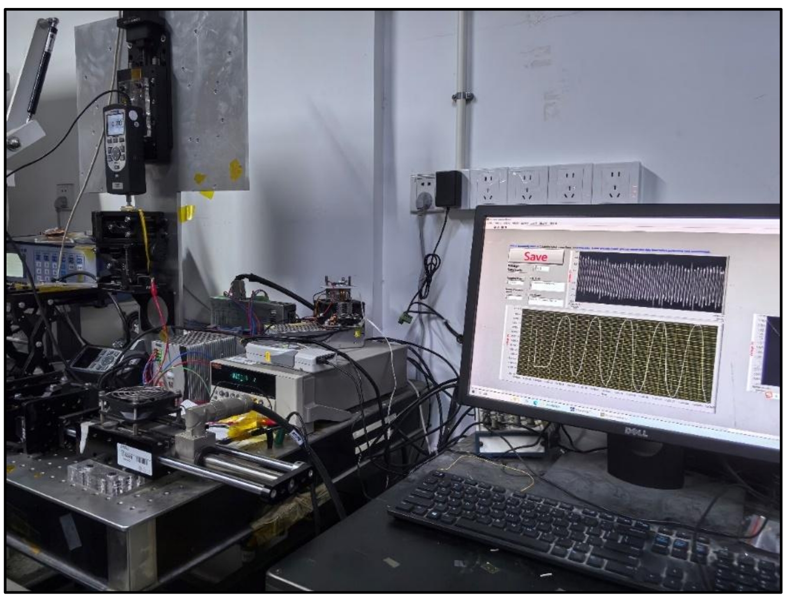


Figure 2.17: LabVIEW interface for real-time monitoring of TENG output signals

By observing the voltage waveforms displayed in LabVIEW in real time, it is possible to assess the stability of the contact process and determine whether there are any significant abnormal fluctuations in the signal.

After the test is completed, the data collected by LabVIEW is imported into OriginPro software for further processing. During the data processing, the unstable data segments at the beginning and end of the test are first removed, and then the stable output signals within the contact-separation cycle are selected as the valid data. For each material, the main extraction is made of its output voltage peak and signal polarity for comparative analysis. By comparing the triboelectric output signals corresponding to different materials, the correspondence relationship between material categories and TENG output characteristics can be established.

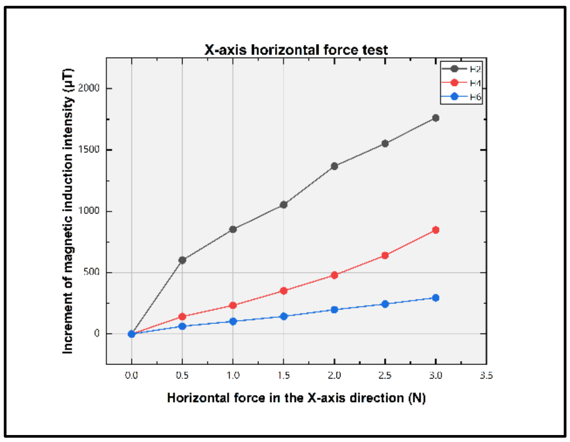
Overall, this test achieved the collection of triboelectric signals from different materials through single-electrode testing method, initial position calibration, controllable contact-separation movements, and controlled variable method. This approach can reduce human operational errors, enhance the comparability of test results for different materials, and provide a reliable data foundation for the subsequent analysis of material identification results.

2.2 Results

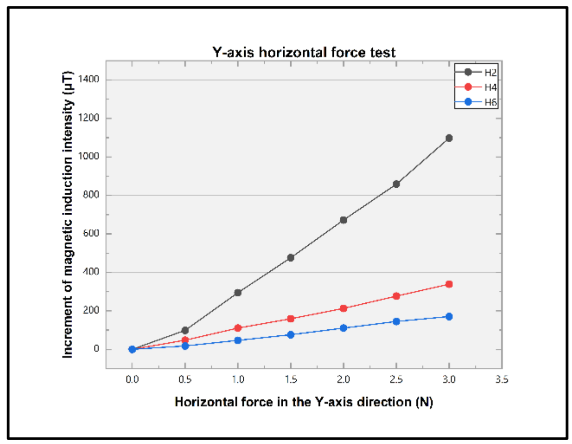
2.2.1 Three-dimensional force test results

Based on the three-dimensional force testing process described in the previous section, this study tested the output response of the three-dimensional force device under the action of external forces. The result is shown in Figure 2.18.

A



B



C

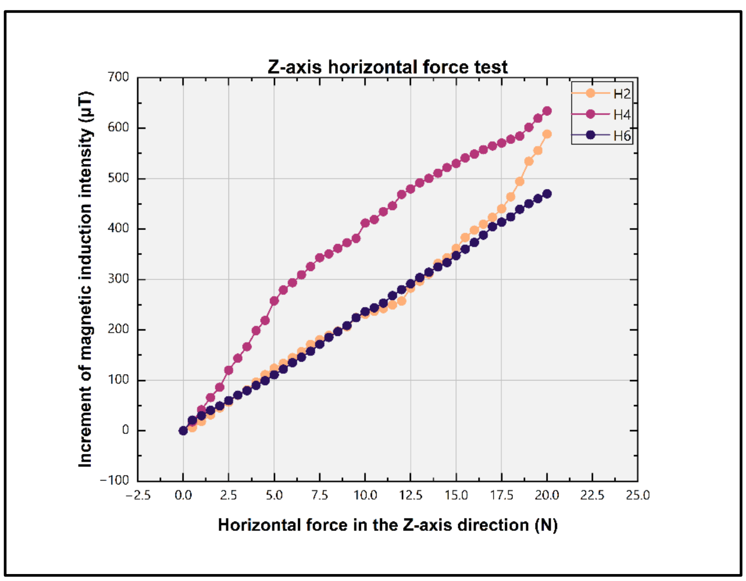


Figure 2.18: Three-dimensional force response of Hall sensing devices: (A) X-axis tangential force; (B) Y-axis tangential force; (C) Z-axis normal force.

It should be noted that in order to further investigate the influence of the distance between the magnet and the Hall sensor chip on the output response, this study fabricated three devices with different magnet placement heights. The magnets were respectively 2 millimeters, 4 millimeters, and 6 millimeters above the MLX90393 chip, and were labeled as H2, H4, and H6. The basic structure and overall fabrication process of the three devices remained consistent, with the only difference being the height of the magnet holes reserved during the PDMS molding process. Therefore, this comparative experiment serves mainly as a supplement to the three-dimensional force test results, and is used to assist in analyzing the impact of different magnet distances on the output response.

As shown in Figure 2.18(A), as the tangential force in the x-direction increases, the magnetic induction intensity increments of the H2, H4, and H6 devices all increase with the increase of the external force, indicating that this structure can effectively respond to the horizontal tangential force in the x-direction. Among them, the output response of the H2 device is the largest, with a magnetic induction intensity increment of approximately 1750μT under a force of 3N, the H4 device is approximately 850μT, and the H6 device is approximately 300μT. This result indicates that under the action of the tangential force in the x-direction, a smaller magnet-chip distance can bring about a greater magnetic field change, thereby achieving a higher output sensitivity.

As shown in Figure 2.18(B), during the process of increasing the tangential force in the y-direction, the increment of the magnetic induction intensity of the three devices also shows a trend of rising with the increase of the applied force. The H2 device still has the highest output response, with an increment of the magnetic induction intensity of approximately 1100μT at 3N; the H4 and H6 devices are approximately 340μT and 170μT respectively. It is worth noting that, compared with the x-direction, the overall output in the y-direction is smaller, indicating that the devices have certain differences in their responses to tangential forces in different directions. This difference may be related to the deformation characteristics of the PDMS pressing layer of the devices in different directions. Due to possible issues such as uneven PDMS curing during the device fabrication process and slight deviations in the installation position of the magnets, the equivalent stiffness of the PDMS layer in different tangential directions may be different. Therefore, under the same magnitude of tangential force, the actual displacement generated by the magnet may be different, thereby causing differences in the magnetic induction intensity changes in the x and y directions.

As shown in Figure 2.18(C), in the normal force test along the z-axis, all three devices exhibited significant magnetic field responses within the range of 0–20N, indicating that the PDMS pressing layer can undergo compression deformation under the normal force and drive the magnet to produce relative displacement. In this direction, the H4 device showed the highest overall response, with the magnetic induction intensity increment reaching approximately 630μT at 20N, the H2 device approximately 590μT, and the H6 device approximately 470μT. Different from the tangential force test, under the normal pressure, it is not the closer the magnet is the greater the output; instead, the H4 device demonstrated a better response. This indicates that for the detection of normal force, an appropriate initial height of the magnet may have better sensitivity.

To further assess the response capabilities of the three devices under forces applied in various directions, this study estimated the average sensitivity based on the relationship between the change in output magnetic flux density and the applied force. It should be noted that the vertical axis in Figure 2.18 represents the change in magnetic induction intensity relative to the initial reference value under different forces:

$$\mathbf{\Delta B = B_F - B_0}$$

Here, B_F denotes the magnetic flux density under a given force, and B_0 denotes the magnetic flux density in the absence of external forces. Therefore, the overall average sensitivity is defined as the ratio of the change in magnetic flux density to the change in force within the range of the maximum applied force:

$$\boldsymbol{S_{avg} = \frac{\Delta B_{max}}{F_{max}}}$$

Here, the sensitivity unit is μT/N. This parameter can reflect the average response capability of the device to changes in force over the entire testing range. Based on the overall changes of each curve in Figure 2.18, the estimated results of the overall average sensitivity of the three devices in the x, y and z directions can be obtained, as shown in Table 1.

Direction	H2 Overall Average Sensitivity (μT/N)	H4 Overall Average Sensitivity (μT/N)	H6 Overall Average Sensitivity (μT/N)
x	587	282	98
y	366	113	57
z	29.4	31.7	23.5

Table 1: Overall average sensitivity

As can be seen from Table 1, there are significant differences in the sensitivity of the devices with different magnet heights in the three-dimensional force detection. For tangential force tests in the x and y directions, the H2 device exhibited the highest sensitivity. This indicates that a smaller gap between the magnet and the device chip enhances the device's magnetic field response to tangential displacement, resulting in a more pronounced change in magnetic flux density for the same magnitude of tangential force. In contrast, the H6 device exhibited the lowest sensitivity in both tangential directions, suggesting that when the magnet is located at a greater distance from the device chip, the magnetic field variation detected at the chip is smaller, thereby reducing the output response.

Furthermore, it can be observed that compared with the tests of tangential forces in the x and y directions, the increment of magnetic induction intensity in the normal force test in the z direction is relatively smaller. This might be related to the deformation mode of the PDMS pressing layer under different force directions. Under the tangential force, the pressing layer is more likely to undergo shear deformation, thereby driving the magnet to produce a more obvious lateral displacement relative to the device chip; while under the normal force, the pressing layer mainly undergoes compression deformation. Due to the good elastic recovery ability of PDMS and the fact that some deformation during vertical pressing may be absorbed by the lateral expansion of the material, it fails to fully convert into an effective relative displacement between the magnet and the device chip. Therefore, the output change in the z direction is smaller than that in the tangential direction, which is reasonable. In this direction, the H4 device exhibits superior overall average sensitivity.

This indicates that for the normal force, the distance between the magnet and the device chip is not necessarily the closer the better. An appropriate distance may be more conducive to achieving a balance between sensitivity and range.

As can be seen from the above results, the fabricated three-dimensional force sensor is capable of producing an output of magnetic flux density increment that varies with the applied force in all three directions (x, y and z), demonstrating that the device possesses three-dimensional force detection capability. Specifically, the tests in the x and y directions verified the device's ability to respond to tangential forces, whilst the test in the z direction verified its detection capability within a range of relatively large normal forces. The comparative results for H2, H4 and H6 further indicate that the mounting height of the magnet affects the magnitude of the output signal; however, this aspect is primarily considered as a supplementary analysis for structural optimization. By comparison, the H2 device demonstrated the best test performance. In summary, the experimental results demonstrate the feasibility of achieving three-dimensional force detection based on Hall sensors and a magnet displacement mechanism.

2.2.2 Material identification test results

Based on the material identification testing process described in the previous section, this study employed a single-electrode contact-separation testing method to collect and analyze the triboelectric output signals generated between different contact materials and the lower electrode plate. The results are shown in Figure 2.19 (The individual voltage signals corresponding to the 13 types of contact materials are provided in Appendix B).

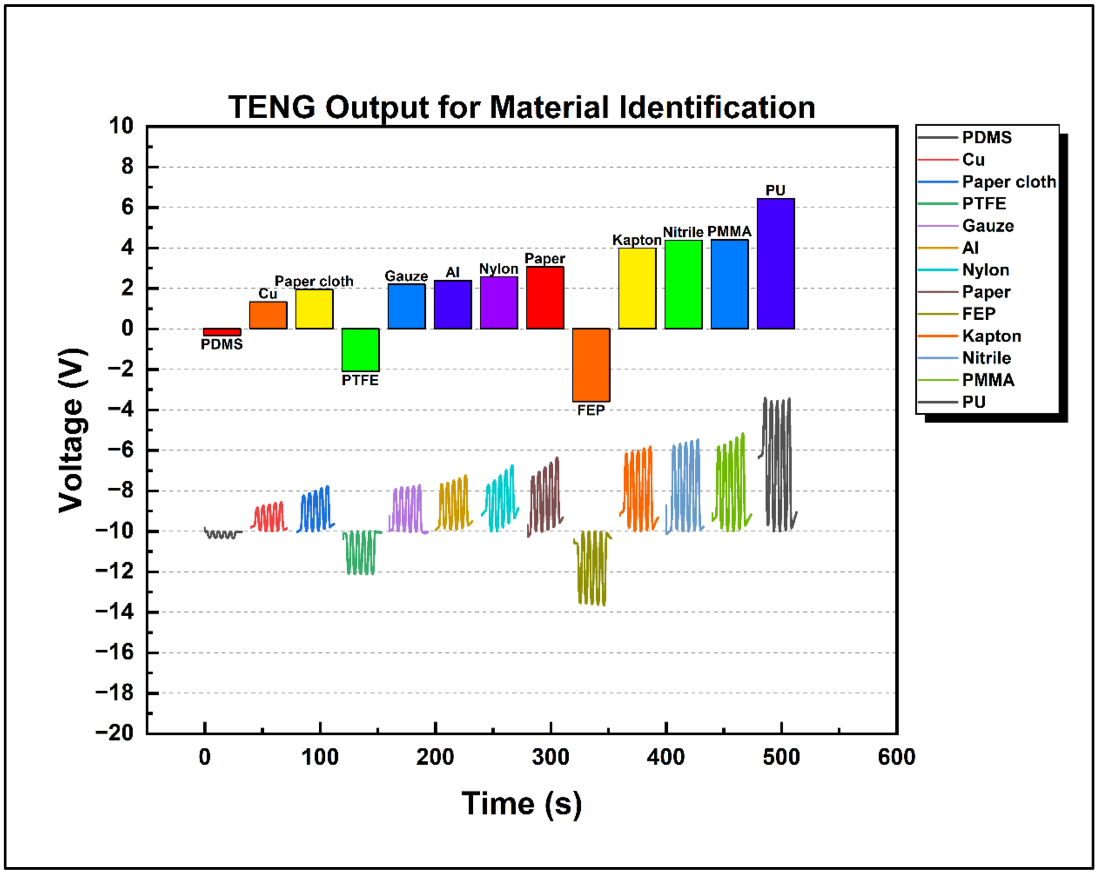


Figure 2.19: Comparison of triboelectric output signals from different materials for TENG material identification

As shown in the figure above, different materials produced different triboelectric output signals under the same testing conditions. Among them, PU generated the highest positive output signal, indicating that under the current testing conditions, PU can generate strong frictional charge transfer between itself and the silicone microstructure layer. Its higher output is related to the triboelectric properties and softness of the material itself. Under the same 5 N force, softer materials may be more likely to form a sufficient contact with the microstructure surface, thereby enhancing the contact charging process.

In contrast, the output signal of PDMS is the smallest. This result is reasonable because PDMS belongs to the silicone material category, and there is a very small difference in the material between it and the silicone contact layer. The net charge transfer during the contact electrification process is very limited, thus resulting in a lower output voltage.

In addition, FEP and PTFE exhibited reverse output signals. This indicates that the direction of charge transfer after they come into contact with the silicone microstructure layer is opposite to that of the material that generates a positive output voltage. Therefore, the polarity of the output signal can be regarded as one of the important characteristics for material identification.

It is worth noting that PMMA and Nitrile produced relatively similar output voltages during the tests, and this phenomenon was consistent in multiple repeated tests. Although PMMA and Nitrile belong to different materials, the output voltage of TENG is not solely determined by the position of the material in the triboelectric sequence. It is also influenced by factors such as the surface roughness, softness, contact stability, and environmental conditions of the materials. Therefore, under the same testing conditions, PMMA and Nitrile are likely to exhibit similar peak voltages due to the similar charge transfer amount. This phenomenon also indicates that when only relying on voltage amplitude and polarity as identification features, there may be a problem of insufficient discrimination between some materials, which could lead to material confusion. Therefore, in subsequent material identification, various features such as the shape of the output signal waveform, peak width, rise and fall time can be further combined to improve the reliability of material identification.

The test results indicate that different materials can generate different triboelectric output signals when in contact with the silicone microstructure layer. These results demonstrate that the TENG device can distinguish different materials based on their voltage amplitude and polarity. Although some materials show similarities under these characteristics, the overall results still confirm the feasibility of the material identification scheme based on TENG.

Overall, this TENG device can collect triboelectric signals from different materials under the same testing conditions, and the results demonstrate a clear ability to distinguish different materials. Combined with the detection results of the three-dimensional force device, the dual-modal tactile sensor proposed in this study can respectively obtain force information through the change in magnetic induction intensity output by the Hall sensor, and obtain material characteristic information through the TENG device, thereby achieving the functions of three-dimensional force detection and material identification.

Chapter 3
Conclusion and Future Work

3.1 Conclusion

This project has designed and fabricated a dual-modal tactile sensing scheme based on Hall sensors and triboelectric nanogenerators (TENG). The main objective is to achieve three-dimensional force detection and material identification functions.

In the three-dimensional force detection section, this study verified the feasibility of the force detection mechanism based on the MLX90393 Hall sensor by establishing the corresponding relationship between the external force and the increment of magnetic induction intensity. The core idea of this scheme is to apply a known magnitude of external force and record the corresponding increment of magnetic induction intensity, thereby establishing a calibration data set consisting of the force and Hall output signal. Based on this data set, the corresponding relationship between the external force and the increment of magnetic induction intensity can be obtained. In the future, if machine learning methods are introduced, this type of data set can be further utilized to train models, achieving more automated and precise three-dimensional force detection.

In addition, this study also conducted comparative tests on devices with different magnet placement heights. The test results of the H2, H4, and H6 devices indicated that the distance between the magnet and the Hall sensor chip would affect the output amplitude and the overall average sensitivity. From the test results, the H2 device demonstrated relatively better sensitivity in the tangential force test, and still maintained a good output response in the normal force test, thus exhibiting relatively superior overall performance.

In the material identification section, this study developed a material identification device based on TENG and used the single-electrode testing method to test and verify various materials. The test results showed that different materials, when in contact with the micro-structured silicone gel contact layer, could generate triboelectric signals with significant differences. The differences mainly manifested in aspects such as voltage amplitude and signal polarity. Among them, PU exhibited a relatively high positive output, FEP and PTFE showed obvious negative outputs, while PDMS's output signal was relatively weak, indicating that the output of TENG is closely related to the triboelectric properties of the contacting materials. Moreover, the output amplitudes of PMMA and Nitrile were relatively close, suggesting that in cases of the same polarity, relying solely on the peak voltage feature may not be sufficient to completely distinguish all materials. Therefore, the test results proved the feasibility of material identification based on TENG, and also indicated that in the future, more voltage features can be combined to further improve the material identification capability.

Overall, this project successfully completed the fabrication and testing of the Hall three-dimensional force detection device and the TENG material recognition device, verifying the feasibility of combining Hall sensors with TENG for dual-modal tactile perception, and providing an experimental basis for subsequent functional integration and array design.

3.2 Future Work

Although this project has verified the feasibility of three-dimensional force detection based on Hall sensors and material identification based on TENG, there are still some aspects that require further improvement and expansion in the subsequent work.

1. Function integration:

In this project, the Hall three-dimensional force device and the TENG material recognition device were fabricated and tested separately. Further work can optimize the device structure and integrate the two functions into a single sensor component, enabling it to simultaneously obtain three-dimensional force information and material recognition signals during a single contact.

2. Optimization of three-dimensional force detection structures:

The test results indicate that the distance between the magnet and the device chip will affect the output amplitude and sensitivity of the device. Therefore, in the future, the installation height of the magnet, the thickness of the PDMS pressing layer, and the structure of the magnetic body hole position can be further optimized to improve the performance of three-dimensional force detection. In addition, the PDMS molding process also needs to be further improved to enhance the uniformity of the PDMS pressing layer and reduce the differences in responses in different directions.

3. Improvement of Material Identification Method:

For material identification, this project has demonstrated that different materials can produce distinct frictional electric output signals. However, the test results also confirmed that relying solely on peak voltage and polarity may not be sufficient for reliable differentiation. Therefore, in the future, more voltage characteristics such as waveform shape, peak width, rise and fall time, and signal area can be further extracted. By combining multiple feature analyses, the accuracy of material identification can be improved.

4. Array-based design:

At present, this project has mainly completed the functional feasibility verification for a single device. After achieving functional integration, it can gradually be expanded to a multi-unit array to realize tactile perception over a wider range, thereby making this sensor more suitable for applications such as robot grasping and human-computer interaction.

In summary, future work will mainly focus on aspects such as functional integration, structural optimization, and improvement in the accuracy of material identification. Through these improvements, the practicality and application potential of this dual-modal tactile sensor scheme have been further enhanced, laying the foundation for its further application in the field of robot tactile perception.

References

[1] H. Q. Huynh, T. Q. Trung, A. Bag, T. D. Do, M. J. Sultan, M. Kim, and N.-E. Lee, "Bio-inspired artificial fast-adaptive and slow-adaptive mechanoreceptors with synapse-like functions," *Adv. Funct. Mater.*, vol. 33, no. 42, Art. no. 2303535, 2023, doi: 10.1002/adfm.202303535.

[2] Y. Gu, T. Zhang, J. Li, C. Zheng, M. Yang, and S. Li, "A new force-decoupling triaxial tactile sensor based on elastic microcones for accurately grasping feedback," *Adv. Intell. Syst.*, vol. 5, no. 3, Art. no. 2200321, 2023, doi: 10.1002/aisy.202200321.

[3] C. Han, Z. Cao, Z. An, Z. Zhang, Z. L. Wang, and Z. Wu, "Multimodal finger-shaped tactile sensor for multi-directional force and material identification," *Adv. Mater.*, vol. 37, no. 19, Art. no. 2414096, 2025, doi: 10.1002/adma.202414096.

[4] Y. Qiu, F. Wang, Z. Zhang, K. Shi, Y. Song, J. Lu, M. Xu, M. Qian, W. Zhang, J. Wu, Z. Zhang, H. Chai, A. Liu, H. Jiang, and H. Wu, "Quantitative softness and texture bimodal haptic sensors for robotic clinical feature identification and intelligent picking," *Sci. Adv.*, vol. 10, no. 30, Art. no. eadp0348, 2024, doi: 10.1126/sciadv.adp0348.

[5] L. Wang, Y. Sun, L. Yang, Y. Sun, Q. Guo, Y. Liu, X. Chen, and Y. Shen, "Touching with torque enables human-level robotic dexterity," *Sci. Adv.*, vol. 12, no. 10, Art. no. eaec3263, 2026, doi: 10.1126/sciadv.aec3263.

[6] X. Qu, Z. Liu, P. Tan, C. Wang, Y. Liu, H. Feng, D. Luo, Z. Li, and Z. L. Wang, "Artificial tactile perception smart finger for material identification based on triboelectric sensing," *Sci. Adv.*, vol. 8, no. 31, Art. no. eabq2521, 2022, doi: 10.1126/sciadv.abq2521.

[7] S. Ali, V. Palaniappan, X. Zhang, D. Maddipatla, B. J. Bazuin, and M. Z. Atashbar, "Investigating the performance of triboelectric nanogenerators (TENGs) fabricated using various flexible polymeric materials," in *Proc. 2022 IEEE Int. Conf. Flexible Printable Sensors Syst. (FLEPS)*, Vienna, Austria, Jul. 2022, doi: 10.1109/FLEPS53764.2022.9781542.

[8] Q. Zheng, J. Lv, J. Zhang, and J. Feng, “Fabrication and application of icephobic silicone coatings on epoxy substrate,” *Prog. Org. Coat.*, vol. 161, Art. no. 106483, 2021, doi: 10.1016/j.porgcoat.2021.106483.

[9] G. Liang, Y. Wang, D. Mei, K. Xi, and Z. Chen, “Flexible capacitive tactile sensor array with truncated pyramids as dielectric layer for three-axis force measurement,” *J. Microelectromech. Syst.*, vol. 24, no. 5, pp. 1510–1519, Oct. 2015, doi: 10.1109/JMEMS.2015.2418095.

[10] S. Niu and Z. L. Wang, “Theoretical systems of triboelectric nanogenerators,” *Nano Energy*, vol. 14, pp. 161–192, 2015, doi: 10.1016/j.nanoen.2014.11.034.

[11] H. Varghese, H. M. Abdul Hakkeem, M. Farman, E. Thouti, S. Pillai, and A. Chandran, “Self-powered flexible triboelectric touch sensor based on micro-pyramidal PDMS films and cellulose acetate nanofibers,” *Results Eng.*, vol. 16, Art. no. 100550, 2022, doi: 10.1016/j.rineng.2022.100550.

Appendix A. Arduino code

Arduino UNO code for Hall sensor data acquisition and SerialPlot monitoring:

```
#include <Wire.h>
#include <Adafruit_MLX90393.h>
#define MLX_ADDR 0x0C
#define DEBUG_MODE 1

Adafruit_MLX90393 mlx = Adafruit_MLX90393();
bool sensor_ok = false;

void setup() {
  Serial.begin(115200);
  while (!Serial) delay(10);

#if DEBUG_MODE
  Serial.println("========================================");
  Serial.println("MLX90393 Single Sensor Init (Robust Mode)");
  Serial.print("I2C Address: 0x"); Serial.println(MLX_ADDR, HEX);
  Serial.println("Config: FILTER=3, OSR=1, GAIN=MLX90393_GAIN_1X");
  Serial.println("Output: Debug info + semicolon-separated 3-column data (X;Y;Z)");
  Serial.println("========================================");
#endif

  Wire.begin();
  Wire.setClock(100000);
  Wire.setWireTimeout(1000, true);

  if (mlx.begin_I2C(MLX_ADDR)) {
```

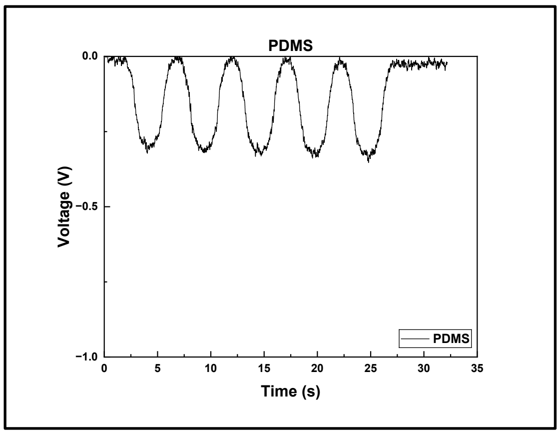
```
    sensor_ok = true;
    mlx.setFilter(MLX90393_FILTER_3);
    mlx.setOversampling(MLX90393_OSR_1);
    mlx.setGain(MLX90393_GAIN_1X);
#if DEBUG_MODE
    Serial.println("Sensor ready.");
#endif
  } else {
#if DEBUG_MODE
    Serial.println("Sensor not found (offline).");
#endif
  }

#if DEBUG_MODE
  Serial.println("========================================");
  Serial.println("Init complete, starting data acquisition...");
#endif

  Serial.println("X;Y;Z");
}

void loop() {
  float x = 0.0, y = 0.0, z = 0.0;

  if (sensor_ok) {
    mlx.readData(&x, &y, &z);
  }

#if DEBUG_MODE
```

```
  Serial.println("-------------------------------------");
  Serial.print("Sensor (0x"); Serial.print(MLX_ADDR, HEX); Serial.print("): ");
  if (sensor_ok) {
    Serial.print("X:"); Serial.print(x, 2); Serial.print(", ");
    Serial.print("Y:"); Serial.print(y, 2); Serial.print(", ");
    Serial.print("Z:"); Serial.print(z, 2); Serial.println();
  } else {
    Serial.println("offline");
  }
#endif

  Serial.print(x, 2); Serial.print(";");
  Serial.print(y, 2); Serial.print(";");
  Serial.println(z, 2);

  Serial.flush();
}
```

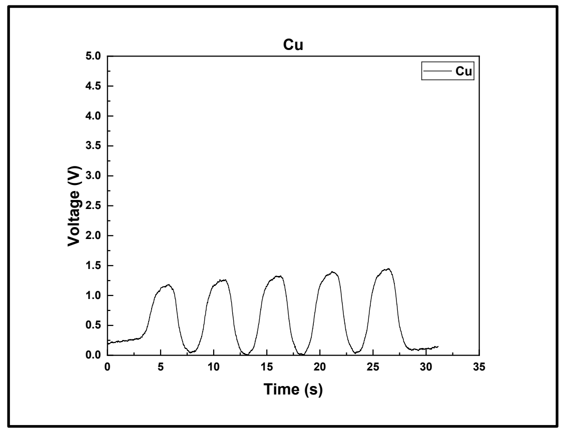
Appendix B. Individual TENG Output Signals for Tested Material

Appendix B presents the TENG output voltage signals corresponding to each test material. All materials were tested under the same single electrode contact-separation test conditions:

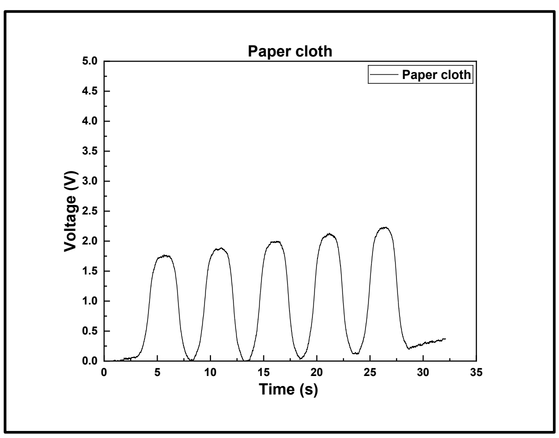
(a) PDMS



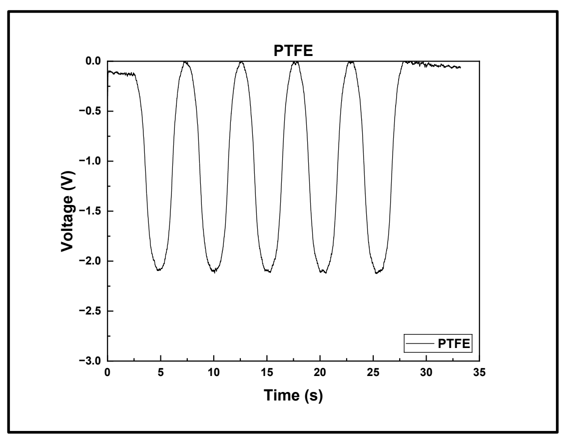
(b) Cu



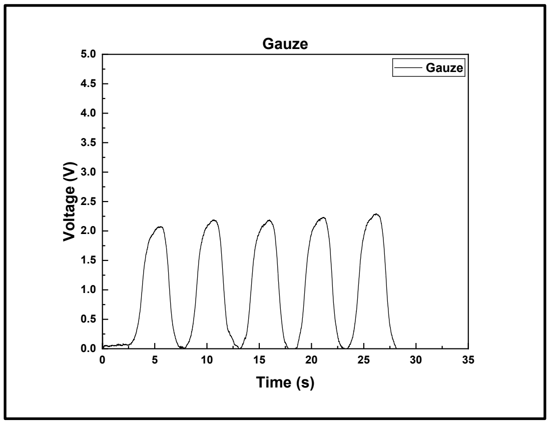
(c) Paper cloth



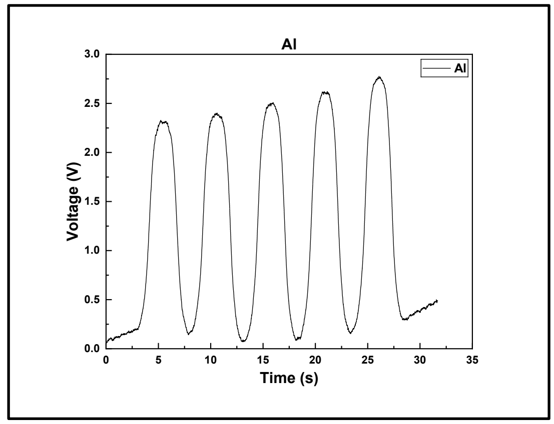
(d) PTFE



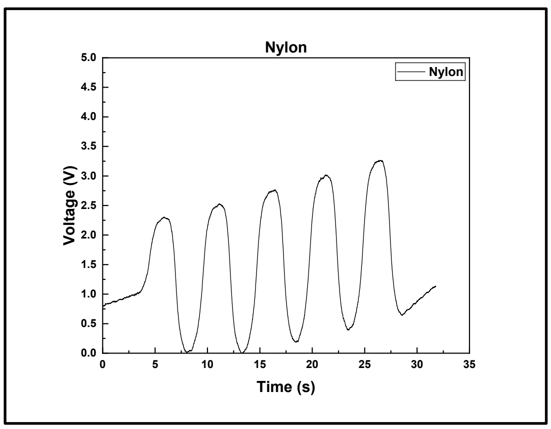
(e) Gauze



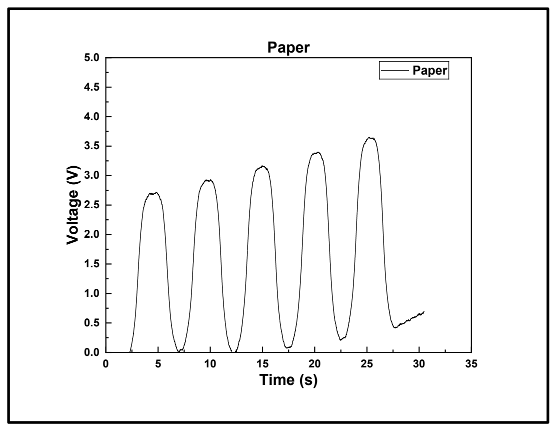
(f) Al



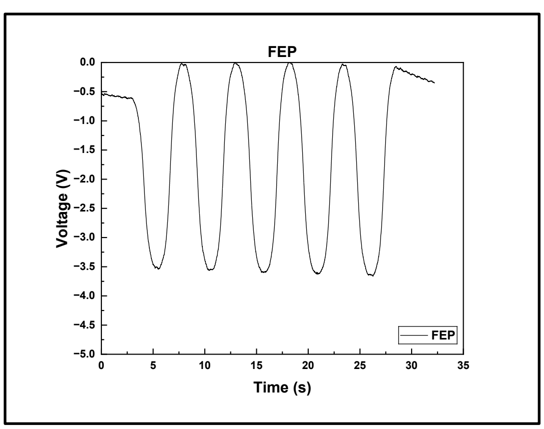
(g) Nylon



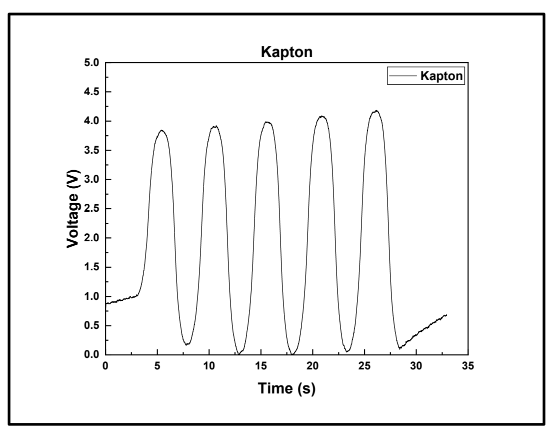
(h) Paper



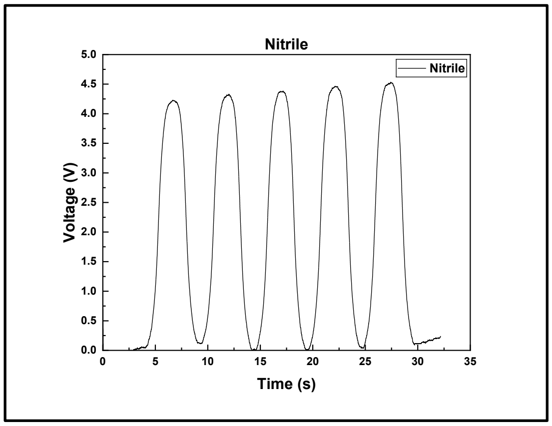
(i) FEP



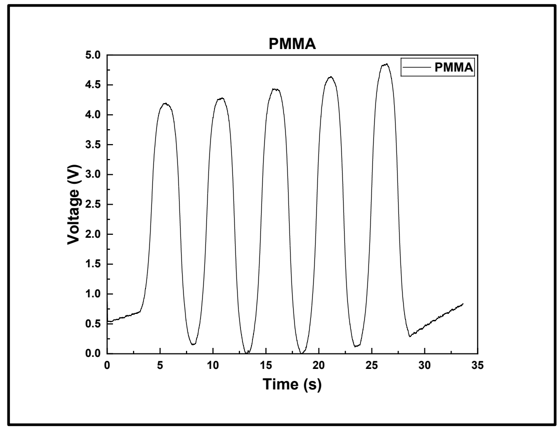
(j) Kapton



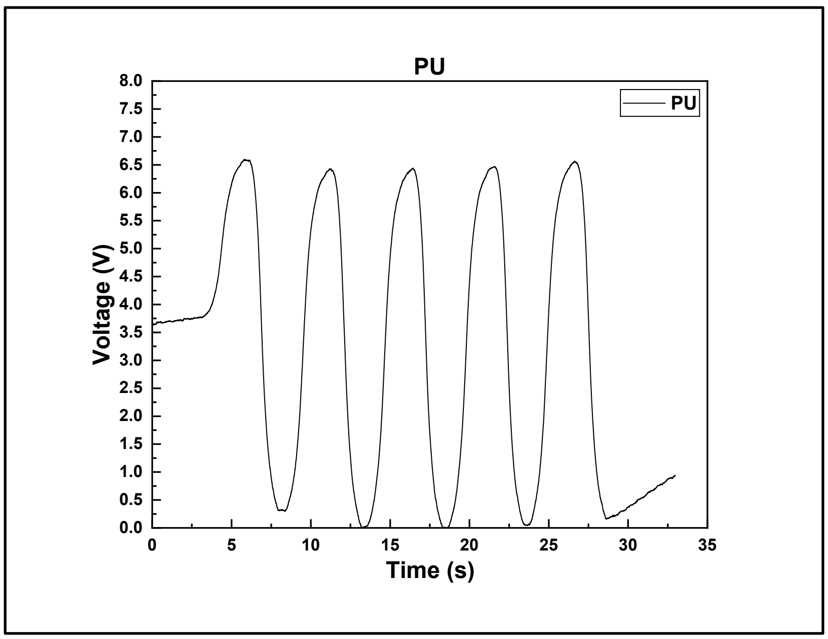
(k) Nitrile



(l) PMMA



(m) PU



Appendix B: Individual TENG output signals of different upper contact materials (a)–(m) under single-electrode contact-separation testing